

# Andrew R. Tawfeek

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## RESEARCH EXPERIENCE

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### Georgia Tech Research Institute

Seattle, WA (Remote)

*Research Scientist*

*May 2025 - Present*

Developing methods for trustworthy AI, spanning topological analysis of neural networks, KV-cache compression, and evaluation of multi-agent LLM systems. Built a Mapper-based pipeline for analyzing approximately 6,900 pathogen genomes and developed filtration algorithms to track neural network topology. Current work evaluates collaboration among AI agents and investigates query-adaptive KV-cache precision for long-context retrieval.

## EDUCATION

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### University of Washington

Seattle, WA

*Doctor of Philosophy in Mathematics*

*Sept. 2021 - June 2026*

**Advisors:** Sándor Kovács, Farbod Shokrieh

**Dissertation:** A Tropical Framework for using Porteous' Formula

### Amherst College

Amherst, MA

*Bachelor of Arts in Mathematics (cum laude)*

*Jan. 2018 - Jan. 2021*

**Advisor:** David A. Cox

## PAPERS AND PREPRINTS

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7. *Taming Wild Knots with Mosaics*, with Mary Deng, Allison K. Henrich, and Sean H. Kawano. Submitted.
6. *The KnotMosaics Package for SageMath*, with Mary Deng, Allison K. Henrich, and Sean H. Kawano. Submitted.
5. *A Tropical Framework for using Porteous' Formula*. Ph.D. Thesis. Under revision.
4. *Topological Signatures of ReLU Neural Network Activation Patterns*, with Vicente Bosca, Tatum Rask, Branden Stone, and Sunia Tanweer. Published in *Proceedings of Machine Learning Research* (2025).
3.  *$A^1$ -Brouwer degrees in Macaulay2*, with Nikita Borisov, Thomas Brazelton, Frenly Espino, Thomas Hagedorn, Zhaobo Han, Jordy Lopez Garcia, Joel Louwsma, and Gabriel Ong. Published in *Journal of Software for Algebra and Geometry* (2024).
2. *On discrete gradient vector fields and Laplacians of simplicial complexes*, with Ivan Contreras. Published in *Annals of Combinatorics* (2023).
1. *Quantum Jacobi forms and sums of tails identities*, with Amanda Folsom, Lizzie Pratt, and Noah Solomon. Published in *Research in Number Theory* (2022).

## HONORS AND AWARDS

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### Amherst Memorial Fellowship ( $\times 3$ )

Amherst College

*Awarded three consecutive years (\$4,300, \$4,600, and \$7,700) for progress in Ph.D.*

*April 2021, 2022, & 2023*

### AMS Sectional Travel Grant

American Mathematical Society

*Awarded for travel to the Spring 2026 Western Sectional Meeting*

*February 2026*

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|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Excellence in Teaching Award</b><br><i>Awarded in recognition of performance and commitment to teaching</i>                                       | University of Washington<br><i>December 2022</i>            |
| <b>Robert H. Breusch Prize in Mathematics</b><br><i>Awarded for, in the opinion of the faculty, presenting the best honors thesis in mathematics</i> | Amherst College<br><i>May 2021</i>                          |
| <b>Inducted Sigma Xi</b><br><i>National scientific research honor society</i>                                                                        | Amherst College<br><i>April 2021</i>                        |
| <b>Provost Fellowship</b><br><i>Awarded an additional \$10,000 towards matriculation to Ph.D. program</i>                                            | University of Washington<br><i>February 2021</i>            |
| <b>Outstanding Poster Award (×2)</b><br><i>Provided a “very clear, well-organized, and enthusiastic presentation on high-level work”</i>             | Joint Mathematics Meeting<br><i>January 2020 &amp; 2021</i> |
| <b>Walker Award for Leadership</b><br><i>Outstanding leadership and contributions to the Mathematics and Statistics community</i>                    | Amherst College<br><i>May 2020</i>                          |

## NOTABLE SOFTWARE PROJECTS

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### **[gemma-4-e4b-it-saes](#)**

Matryoshka batch-topk sparse autoencoders for the instruction-tuned **[gemma-4-E4B-it](#)**, released on HuggingFace in **[sae\\_lens](#)** format after the official base-model SAEs were found to fail on its activations.

### **[KnotMosaics](#)**

SageMath package for constructing, validating, and analyzing knot mosaic diagrams, with strand tracing, planar diagram codes, and rational tangle mosaics; submitted upstream to SageMath.

### **[neural-network-elements](#)**

Interactive web-based tool for building, training, and analyzing multi-layer ReLU neural networks with real-time decision boundary visualization, polyhedral decomposition, and dual graph spectral analysis. Developed at GTRI.

### **[M2-engine](#)**

Developer-facing tooling for the Macaulay2 C++ engine: an interactive subsystem explorer, visual guide, and Doxygen API documentation, built for the *Building the Future of Macaulay2* workshop at Georgia Tech.

### **[steering](#)**

Local interpretability workbench for live SAE feature steering in TransformerLens models (FastAPI UI, CLI, and TUI), built at OpenAI’s Codex Hackathon.

### **[volatility-complex](#)**

Persistent-homology pipeline detecting S&P 500 regime shifts from sector-volatility correlations; capstone for the Erdős Institute Quantitative Finance Bootcamp.

### **[discrete-gradients](#)**

Python implementation of discrete gradient vector fields and Laplacian eigenvalue computation on simplicial complexes, accompanying published research in *Annals of Combinatorics*.

## TEACHING

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|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>University of Washington</b><br><i>Instructor</i>                                                                                          | Seattle, WA<br><i>Summer 2023</i>           |
| – <b>MATH 441: Topology:</b> Introduction to metric and topological spaces, convergence, continuity, products, connectedness, and compactness |                                             |
| <b>University of Washington</b><br><i>Teaching Assistant</i>                                                                                  | Seattle, WA<br><i>Sep. 2021 - June 2026</i> |
| – <b>MATH 464/465: Numerical Analysis I &amp; II:</b> Sp2024, Au2024, Wi2025, Au2025, Wi2026                                                  |                                             |
| – <b>MATH 442/443: Differential Geometry of Curves and Surfaces:</b> Winter 2023, Spring 2023                                                 |                                             |
| – <b>MATH 300: Introduction to Mathematical Reasoning:</b> Spring 2026                                                                        |                                             |
| – <b>MATH 224: Advanced Multivariable Calculus:</b> Spring 2025                                                                               |                                             |

- **MATH 208: Linear Algebra with Applications:** Summer 2022
- **MATH 180/ART 255: Art and Mathematics as Embodied Practices:** Autumn 2022
- **MATH 125: Calculus with Analytic Geometry II:** Au2021, Wi2022, Sp2022, Au2023, Wi2023, Wi2024

## University of Washington

Seattle, WA

*Washington Directed Reading Program (WDRP)*

*Jan. 2022 - June 2026*

- **Role:** Mentored undergraduate students on quarter-long independent studies on advanced topics
- **Topics:** Discrete Morse Theory (Wi2022), Persistent Homology (Wi2022), Chip Firing on Graphs (Sp2022), Nonlinear Algebra (Au2022), Singularities (Sp2024), Knot Theory (Au2024), Ideals, Varieties, and Algorithms (Wi2025)

## University of Washington

Seattle, WA

*Washington eXperimental Mathematics Lab (WXML)*

*Sep. 2022 - June 2026*

- **Permutation Polynomials:** Studied bijective polynomial automorphisms of finite fields and their correspondence with elements of the symmetric group (Au2023, Wi2023)
- **Gerrymandering:** Developed fair redistricted maps of Washington state via Markov chains, Monte Carlo methods, and generative AI (Sp2024)
- **Teaching a Computer to Knot:** Built and trained a graph neural network to classify knots and knot invariants using matrix-structured mosaics (Au2025)
- **Wild Knot Mosaics:** Developed an expanded theory of mosaics to represent wild knots exhibiting infinite-complexity along their strands (Wi2025–2026)

## Amherst College

Amherst, MA

*Course Grader*

*Aug. 2020 - May 2021*

- **Fall 2020:** MATH 460: Analytic Number Theory, MATH 255: Geometry
- **Spring 2021:** MATH 420: Arithmetic of Elliptic Curves, MATH 310: Theory of Partitions

## Bristol Community College

Fall River, MA

*Supplemental Instructor*

*Jan. 2017 - May 2017*

- **Multivariate Calculus:** Met students three times per week (twice in person, once online) to review material and reinforce concepts

## Bristol Community College

Attleboro, MA

*Subject Tutor*

*Sept. 2016 - Dec. 2017*

- **Subjects:** Statistics, algebra, precalculus, calculus, differential equations, classical mechanics, electromagnetism, and general chemistry; met with 10–15 students weekly

## SERVICE

### Reviewer for zbMATH Open

FIZ Karlsruhe

*Providing reviews of published articles and books in various fields of mathematics*

*Nov. 2024 - Present*

### Developer for Macaulay2 and SageMath

Open Source

*Contributor to the computer algebra system and its C++ engine internals*

*2023 - Present*

### Delegate for Mathematical Sciences on Capitol Hill (#MathSciOnTheHill)

Washington, D.C.

*One of 280 mathematicians; advocated for research funding in U.S. House and Senate meetings*

*Jan. 2026*

### Organizer of 5+ Graduate Reading Seminars

University of Washington

*Topics: algebraic geometry, tropical geometry, knot theory,  $\infty$ -categories, arithmetic topology*

*2022 – 2025*

### Joint Mathematics Meeting Fundraising (\$10,850 for 14 students)

Amherst College

*Acquired funding after budgetary work in-discussion with administration and college president's office*

*2019 & 2020*